Lecture 10: Blackboard Notes (Mechanical Properties of Active Materials)

A suggested sequence of boards. Each block lists what to write; arrows show the logical flow, sketches to draw are noted in italics. Symbols are defined as they first appear. Where a slide from your deck supports a board it is flagged "Show Slide N" – slides 20-28 accompany Boards 1-3, 5, and 7.

What this lecture is about

An equilibrium material has well-defined mechanical properties – a stress set by external load, an elasticity that stores energy, a pressure fixed by the bulk – all following from thermodynamics. **Inject energy at the microscale and every one of these gets modified.** Guiding question: *what happens to stress, fluctuations, elasticity, pressure, and rigidity when the material makes its own forces?*

The thread. We walk through the standard mechanical observables one by one and ask how activity changes each: internal stress (Board 1-2), fluctuations (Board 3-4), elasticity (Board 5), pressure (Board 6), and rigidity (Board 7-8).

Surprises compared to equilibrium.

- An active material carries a **built-in stress** that drives flow with no external load; activity can even make the **viscosity negative**.
- Fluctuations **violate the fluctuation-dissipation theorem** – you can have large fluctuations without matching dissipation.
- Elasticity can become **odd** (break time-reversal): a closed deformation cycle does net work – a tiny motor.
- Pressure stops being a **state function** and drives phase separation with no attraction.
- Rigidity becomes **tunable by activity**, not just by composition.

Board 1: Active stress – where the internal force comes from

- microscopically, each active unit is a **force dipole** (pusher/puller, recall Lecture 5; the nematic version $\sigma = -\zeta \mathbf{Q}$ from Lecture 8)
- microscopic model = the **active Brownian particle** (ABP, Lecture 3): self-propels at speed v_0 along $\mathbf{n}$, whose direction diffuses:

$$\dot{\mathbf{r}} = v_0\,\mathbf{n} + \sqrt{2D_T}\,\eta(t), \qquad \dot{\mathbf{n}} = \sqrt{2D_R}\,\xi(t) \times \mathbf{n}$$

persistence length $\ell_p = v_0/D_R$ (D_T, D_R = translational / rotational diffusion) - coarse-grain a distribution $c(\mathbf{r}, \mathbf{n})$ of active units (position $\mathbf{r}$, orientation $\mathbf{n}$) $\Rightarrow$ macroscopic **active stress tensor**:

$$\sigma_{ij}^{\text{active}} = -\,\zeta\Delta\mu \int n_i n_j\, c(\mathbf{r}, \mathbf{n})\, d\mathbf{n} \;=\; \alpha\, c\, Q_{ij}$$

- $\zeta\Delta\mu$ = **self-propulsion strength**: $\Delta\mu$ = chemical fuel (e.g. ATP) per turnover, ζ converts it to force; Q_{ij} = nematic order tensor

- **isotropic limit** (no orientational order, $\mathbf{Q} = 0$): a uniform active pressure $\sigma_{ij}^{\text{active}} = -\zeta \Delta\mu \, c \, \delta_{ij}$
- key point: this stress is generated **internally by energy conversion**, not by an external load
- **surprise**: in equilibrium a stress needs an applied load or thermal motion; an active material carries a stress that drives flow on its own

Show Slide 20 ("Creating Active Stresses"): the chain of force dipoles and their superposed flow field.

Board 2: Force transmission – how the stress spreads

- **momentum balance** – force per volume = divergence of the stress (only a *varying* stress pushes; a uniform one pushes equally on opposite faces of a blob ⇒ no net force):

$$\nabla \cdot (\sigma^{\text{passive}} + \sigma^{\text{active}}) + \mathbf{f}^{\text{ext}} = \rho \, \partial_t^2 \mathbf{u}$$

($\mathbf{u}$ = **displacement**, so $\partial_t^2 \mathbf{u}$ = acceleration – *not* the velocity/flow field. At the microscale inertia is negligible ⇒ force balance $\nabla \cdot \sigma + \mathbf{f} = 0$; keeping the $\rho \, \partial_t^2 \mathbf{u}$ term is what yields elastic / active **waves**.)

- so: one active unit pushes here – **how far does the disturbance reach?** Active units are force **dipoles** (no net force), so the *transmitted stress* falls off as a power law set by the medium and the dimension. The ladder in 3D (each step = one spatial derivative, costing one power of r):

$$\underbrace{1/r}_{\text{point-force displ.}} \xrightarrow{\text{dipole}} \underbrace{1/r^2}_{\text{dipole displ.}} \xrightarrow{\text{stress}} \underbrace{1/r^3}_{\text{transmitted stress}} \qquad (\text{in 2D: } \log r \to 1/r \to 1/r^2)$$

- **viscous fluid** and **elastic network**: power-law ⇒ **long-ranged**; a **jammed / granular** packing screens **exponentially** (forces travel only along "force chains") ⇒ very short-ranged
- **screening length** – a homogeneous fluid or solid does *not* screen force (the power-law decay above has no intrinsic length); a finite range needs a momentum sink, a friction coefficient Γ (drag per volume to the solvent or a substrate):

$$\ell_{\text{screen}} = \sqrt{\eta/\Gamma}, \qquad \ell_{\text{screen}}^{\text{active}} = \sqrt{(\eta + \eta_{\text{active}})/\Gamma}$$

activity adds to the effective viscosity ⇒ a **longer** screening length (forces reach farther). *(The ratio η/G is instead the Maxwell relaxation* time τ*, not a length.)*

- **bacterial superfluidity**: shear a suspension of pushers and the effective viscosity is

$$\eta_{\text{eff}} = \eta_0 \, (1 - \alpha \, n \, v_0) \qquad (n = \text{density}, \ v_0 = \text{swim speed})$$

- *why it drops*: the shear tilts the swimmers; a pusher's outward (extensile) flow then pushes fluid the **same way** as the shear and does part of the work ⇒ less external stress is needed to keep it flowing
- with enough pushers they do **all** the work: $\eta_{\text{eff}} \to 0$ (superfluid), then **negative** (the bacteria drive the rheometer instead of resisting it)
- **surprise**: activity can make the viscosity vanish or go negative – disturbances are *amplified*, not damped, so forces reach much farther than in any passive medium

Show Slide 21 ("Superfluidity of Bacterial Suspensions", López et al. 2015): the effective viscosity drops well below the solvent's.

Board 3: Fluctuations – effective temperature and broken FDT

- equilibrium: thermal fluctuations are Gaussian and obey the **fluctuation-dissipation theorem (FDT)** – the size of fluctuations is locked to the dissipation (response):

$$\text{fluctuations} \;\propto\; k_B T \times (\text{dissipative response})$$

- active: fluctuations are **non-Gaussian** and **violate the FDT**. Define an **effective temperature** from the ratio

$$T_{\text{eff}} = \frac{\text{fluctuation}}{\text{response}} \;>\; T$$

- concretely, for a normal mode of stiffness k_n and amplitude q_n, equilibrium **equipartition** $\langle q_n^2 \rangle = k_B T / k_n$ becomes

$$\langle q_n^2 \rangle = \frac{k_B T_{\text{eff}}(n)}{k_n}$$

  with $T_{\text{eff}}(n)$ depending on the mode n – the giveaway that it is not a true temperature
- transport: enhanced and even **superdiffusive** at short/intermediate times, $\langle |\Delta u(r,t)|^2 \rangle \sim t^\alpha,\ \alpha > 1$
- **surprise**: T_{eff} is *not* a real temperature – different observables give different T_{eff}; the system is genuinely out of equilibrium
- **how you actually see it** (Mizuno): measure two things *independently* – the **response** (microrheology: push a probe bead with a known force, record its displacement $\Rightarrow \chi''$) and the **spontaneous fluctuations** (just track the same bead). Equilibrium locks the two together by the FDT; with motors on the fluctuations **overshoot** the FDT prediction by orders of magnitude at low ω; remove the ATP and the FDT is **restored exactly**

Show Slides 22-23 (Turlier et al. 2016, red-blood-cell flickering): compare the fluctuations $C(f)$ with the response $\chi''(f)$ – they coincide with ATP depleted (FDT holds) but split apart in fresh cells (FDT broken), giving an effective energy up to $\sim 25\, k_B T$. Also Mizuno et al. 2007, actin-myosin gel.

Board 4: Power spectral density and the Kramers-Kronig relations

- power spectral density $S_x(\omega)$ = how fluctuation power is distributed over frequency
- **passive** part (FDT): a Lorentzian

$$S_x^{\text{eq}}(\omega) = \frac{2 k_B T\, \tau}{1 + (\omega\tau)^2}$$

  - **plateau** at low ω = the particle is **confined** (elastic restoring force G_0, relaxation time τ); $1/\omega^2$ tail = viscous
- **active** part: an extra low-frequency excess $S_x^{\text{active}} \sim A/\omega^\beta \;\Rightarrow$ **enhanced fluctuations at low frequency**
- response $\chi(\omega) = \chi'(\omega) + i\chi''(\omega)$: χ' = in-phase **storage** (elastic), χ'' = out-of-phase **loss** (dissipation)
  - everyday feel: **rubber band** = almost all storage; **honey** = almost all loss; **silly putty** = both, set by *how fast* you push it (bounces when thrown = fast/elastic, slumps when left = slow/viscous)
- **(if asked) Kramers-Kronig**: causality ($\chi(t<0) = 0$ – no response before the kick) forces storage and loss to be **Hilbert transforms** of each other,

$$\chi'(\omega) = \frac{1}{\pi}\, \mathcal{P}\!\int \frac{\chi''(\omega')}{\omega' - \omega}\, d\omega'$$

so they are **not independent**: know one at all ω, get the other – causality's universal *accounting*, where how a material stores energy at one frequency fixes how it must dissipate at all others. This holds for **any** causal linear material – passive *or* active - in rheology one uses the complex **modulus** $G^*(\omega) = G'(\omega) + iG''(\omega)$ (storage G', loss G''; the response/compliance is $\chi = 1/G^*$). The same causality gives the modulus KK pair:

$$G'(\omega) = \frac{2}{\pi}\,\mathcal{P}\!\int_0^\infty \frac{\omega' G''(\omega')}{\omega'^2 - \omega^2}\, d\omega', \qquad G''(\omega) = -\frac{2\omega}{\pi}\,\mathcal{P}\!\int_0^\infty \frac{G'(\omega')}{\omega'^2 - \omega^2}\, d\omega'$$

- **FDT** (equilibrium only): the fluctuations are fixed by the loss, $S_x(\omega) = \dfrac{2k_BT}{\omega}\,\chi''(\omega) = \dfrac{2k_BT}{\omega}\,\mathrm{Im}[1/G^*(\omega)]$
- so in equilibrium one measurement of χ'' gives both the storage (via KK) **and** the fluctuations (via FDT) – all tied together
- **full spectrum** = equilibrium (FDT) part + active excess:

$$S_x(\omega) = S_x^{\mathrm{eq}}(\omega) + S_x^{\mathrm{active}}(\omega) = \frac{2k_BT}{\omega}\,\mathrm{Im}[1/G^*(\omega)] + \frac{A}{\omega^\beta}$$

(the simple model at the top of the board takes S_x^{eq} as the Lorentzian) - **why activity can break this** – *time-reversal*: run a film of a passive material backwards and it looks fine (a pendulum swings the same either way); run a film of swimming bacteria backwards and they swim backwards, which never happens. Energy flowing one way **breaks time-reversal symmetry**, and that is what lets the fluctuation $\leftrightarrow$ dissipation link fail - active: **causality (KK) still holds**, but the **FDT bridge breaks** – the measured S_x **exceeds** $\frac{2k_BT}{\omega}\chi''$ (exactly the Mizuno/Turlier gap of Board 3); that excess is the non-equilibrium signature - **surprise**: in equilibrium you cannot have fluctuations without matching dissipation; activity breaks that link (the FDT, not causality)

Sketch: log-log S_x vs ω – passive plateau then $1/\omega^2$; active curve lifted at low ω.

Board 5: Elasticity tensor and odd elasticity

- Hooke's law (small strain): $\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}$, strain $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ ($\mathbf{u}$ = displacement)
- σ and ε symmetric ($\sigma_{ij} = \sigma_{ji}$, $\varepsilon_{ij} = \varepsilon_{ji}$) $\Rightarrow$ **minor symmetries** $C_{ijkl} = C_{jikl} = C_{ijlk}$
- equilibrium: stress derives from a stored-energy density w, $\sigma_{ij} = \partial w/\partial\varepsilon_{ij} \Rightarrow C_{ijkl} = \partial^2 w/\partial\varepsilon_{ij}\partial\varepsilon_{kl} \Rightarrow$ **major symmetry** $C_{ijkl} = C_{klij}$
- isotropic passive solid (bulk modulus K, shear modulus G): $C_{ijkl} = K\,\delta_{ij}\delta_{kl} + G(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{3}\delta_{ij}\delta_{kl})$
- **odd elasticity** – the simple picture:
  - a **normal spring**: push it, it pushes **straight back** ($F = -kx$); it stores energy and gives it back, so squeeze-and-release around a loop does **zero** net work
  - an **odd-elastic solid**: push down and it also **shears sideways**, and that sideways response is **non-reciprocal** $\Rightarrow$ going **around a deformation loop the material does net work** (a tiny engine)
- why a passive solid cannot: its force comes from a stored energy (a potential), forcing C symmetric. The extra "odd" part has **no energy to store**:

$$C_{ijkl}^{\mathrm{antisym}} = \tfrac{1}{2}(C_{ijkl} - C_{klij}) \neq 0, \qquad W = \oint \sigma_{ij}\, d\varepsilon_{ij} = C_{ijkl}^{\mathrm{antisym}} \oint \varepsilon_{kl}\, d\varepsilon_{ij} \neq 0$$

so the work over a cycle must be **fed by activity**, and the response **breaks time-reversal** (run the film backwards and it looks wrong) - a concrete handle (2D): in Voigt notation C becomes a small matrix C_{ij} and the odd part is $C_{ij}^{\mathrm{odd}} = \frac{1}{2}(C_{ij} - C_{ji})$; the **odd modulus** K^o couples a **compression** to an internal **torque / transverse shear** – push straight in, get a sideways response (a passive, symmetric C cannot). It is the *static* cousin of a gyroscopic or Lorentz force, but driven by **deformation** instead of velocity - it needs **broken parity** (chirality – units that all spin the same way) *and* energy input: exactly what the spinning starfish embryos provide - **surprise**: equilibrium elasticity only *stores* energy; odd elasticity *turns activity into mechanical work* – a solid that is also a motor

Sketch: normal spring (force back along the push) vs odd spring (force at a right angle); a stress-strain loop whose enclosed area = work per cycle.

Show Slides 24-27: Scheibner et al. (2020) schematic (push → non-reciprocal shear), then Tan et al. (2022) living chiral crystals of spinning starfish embryos – the measured odd-vs-passive shear response.

Board 6: Swim pressure and the active equation of state

- a self-propelled particle pushes on its confining walls ⇒ a **swim pressure**:

$$P_{\text{swim}} = n\,\zeta\,\frac{v_0^2}{2dD_r} = n\,\zeta\,D_A, \qquad D_A = \frac{v_0^2}{2dD_r}$$

  n = density, ζ = drag coefficient, v_0 = swim speed, D_r = rotational diffusion, d = dimension, D_A = active diffusivity *(note: this ζ is the drag, not the activity ζ of Board 1)*

- *where it comes from*: a swimmer runs into a wall and pushes on it for about one **persistence time* * $1/D_r$ (until its heading randomizes), then leaves; averaging force × time over many such events gives the pressure
- **analogy**: $P_{\text{swim}} = n\,(\zeta D_A)$ is the *ideal-gas* pressure of active particles, with the **active energy** ζD_A replacing $k_B T$; the persistence length $\ell_p = v_0/D_r$ sets the scale – more persistent ⇒ larger swim pressure
- total: $P = P_{\text{ideal}} + P_{\text{swim}} + P_{\text{interaction}}$
- **MIPS link**: crowding **slows** the swimmers, $v(n)\downarrow$, so $P(n)$ turns **non-monotonic** ($\partial P/\partial n < 0$) ⇒ spinodal ⇒ phase separation with **no attraction** (Lecture 6)
- *not a state function*: unlike an equilibrium pressure, P_{swim} can depend on the **confining geometry**; in general $P = P(n, v_0, D_r, \text{interactions})$ rather than $P(n,T)$ alone – only torque-free spheres give a true state function (Solon et al. 2015)
- **surprise**: equilibrium pressure is a state function set by the bulk; swim pressure can depend on the boundary and drives phase separation with no attractive forces

Board 7: Activity-induced stiffening and softening

- activity can **stiffen**: motor prestress σ in a semiflexible network raises the modulus

$$K(\sigma) = \frac{d\sigma}{d\gamma} \sim \sigma^{3/2} \qquad \text{(MacKintosh)} \qquad \Rightarrow \quad K_{\text{eff}} \sim (\sigma_{\text{ext}} + \sigma_a)^{3/2}$$

  ($K = d\sigma/d\gamma$ = differential modulus, γ = shear strain; σ_a = active prestress); actin-myosin: up to 100× stiffer with motors on
- activity can **soften**: active forces help particles escape cages ⇒ **unjamming**. Activity a shifts the jamming density and lowers the yield stress:

$$\phi_J^{\text{eff}} = \phi_J^0 + f(a, \nabla\rho), \qquad \sigma_y^{\text{eff}} = \sigma_y^0 - \zeta a$$

- so activity is a **two-way knob** on rigidity
- **surprise**: a material's stiffness becomes tunable by **internal energy input**, not just by composition or temperature

Show Slide 28 ("Stiffening of Active Materials", Koenderink et al. 2009): K' vs applied stress with the $\sim \sigma^{3/2}$ collapse; switching motors on stiffens the network up to $100\times$.

Board 8: Critical phenomena in active materials

- activity-driven mechanical transitions (**unjamming**, **rigidity percolation**) follow the usual critical scaling, with **activity a as the control parameter**:

$$\psi \sim |a - a_c|^{\beta}, \qquad \xi \sim |a - a_c|^{-\nu}, \qquad \tau \sim |a - a_c|^{-z\nu}$$

(ψ = order parameter, e.g. yield strain; ξ = correlation length; τ = relaxation time; a_c = critical activity) - concrete cases: **rigidity-percolation** modulus $G \sim (c - c_p(c_a))^f$ (c = connectivity, threshold c_p shifted by activity c_a, f = rigidity exponent); active **shear-band** width $w \sim a^{-1/2}$ - same framework as the equilibrium critical points of Lecture 6, and the flocking transition of Lecture 9 – now driven by activity - **surprise**: activity plays the role temperature or density plays in equilibrium – it is the knob that tunes the material through a mechanical phase transition